

# UNDER THE INFLUENCE: QUANTIFYING PERSUASION AND VIGILANCE IN LARGE LANGUAGE MODELS

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## ABSTRACT

With increasing integration of Large Language Models (LLMs) into high-stakes decision-making, understanding their social capacities—*vigilance* (the ability to resist misleading input) and *persuasion* (the ability to influence others)—is critical for AI safety. We introduce a controlled evaluation framework based on the Sokoban puzzle game to jointly measure these capacities in five frontier LLMs: GPT-5, Grok 4 Fast, Gemini 2.5 Pro, Claude Sonnet 4, and DeepSeek-R1. We provide formal metric definitions for performance, persuasion, and vigilance, and find that these three capacities are dissociable: strong task performance neither guarantees persuasive ability nor resistance to malicious advice. We further show that token usage patterns are resource-rational—models spend more tokens resisting deception, even when they fail to do so. To address concerns about generalizability, we introduce a **cross-domain trivia persuasion task** showing that persuasion/vigilance dissociation holds outside Sokoban, and a **vigilance prompting ablation** demonstrating that simple prompting strategies (chain-of-thought, skeptic framing, advice verification) improve vigilance by 8–23% relative to baseline. Our work provides the first framework that *jointly* quantifies performance, persuasion, and vigilance in a long-horizon sequential puzzle game with algorithmically controlled advice quality, offering formal, independently measurable metrics and actionable guidance for improving LLM resistance to deception.

## 1 INTRODUCTION

Large Language Models (LLMs) are rapidly being integrated into many aspects of daily life as “thought partners” (??), assisting humans with tasks ranging from selecting restaurants to allocating investments. As these systems operate more autonomously, they must interface with large volumes of information—some benevolent, some intentionally malicious—and distill it into actionable guidance.

This creates two critical AI safety risks. First, LLMs may be susceptible to malicious external content (????), which they may unwittingly relay to users. Second, LLMs themselves may have misaligned objectives and *persuade* users toward sub-optimal or harmful decisions (?). Both risks hinge on two *social capacities*: *vigilance*—the ability to detect and resist manipulation—and *persuasion*—the ability to construct compelling arguments. Prior work has studied persuasion and vigilance largely separately (???), and while recent work jointly addresses both in game settings (??), the relationship between these capacities and basic task competence— and formal, independently measurable metrics for all three— remain understudied.

We address this gap by introducing an evaluation framework based on Sokoban, where one LLM acts as an “advisor” providing strategy advice (either benevolent or malicious) to another LLM “player” solving the puzzle. This design allows us to simultaneously measure task performance, persuasion, and vigilance under controlled conditions. We evaluate five frontier models and find that these capacities are *dissociable*: strong unassisted performance does not imply either persuasive ability or vigilance.

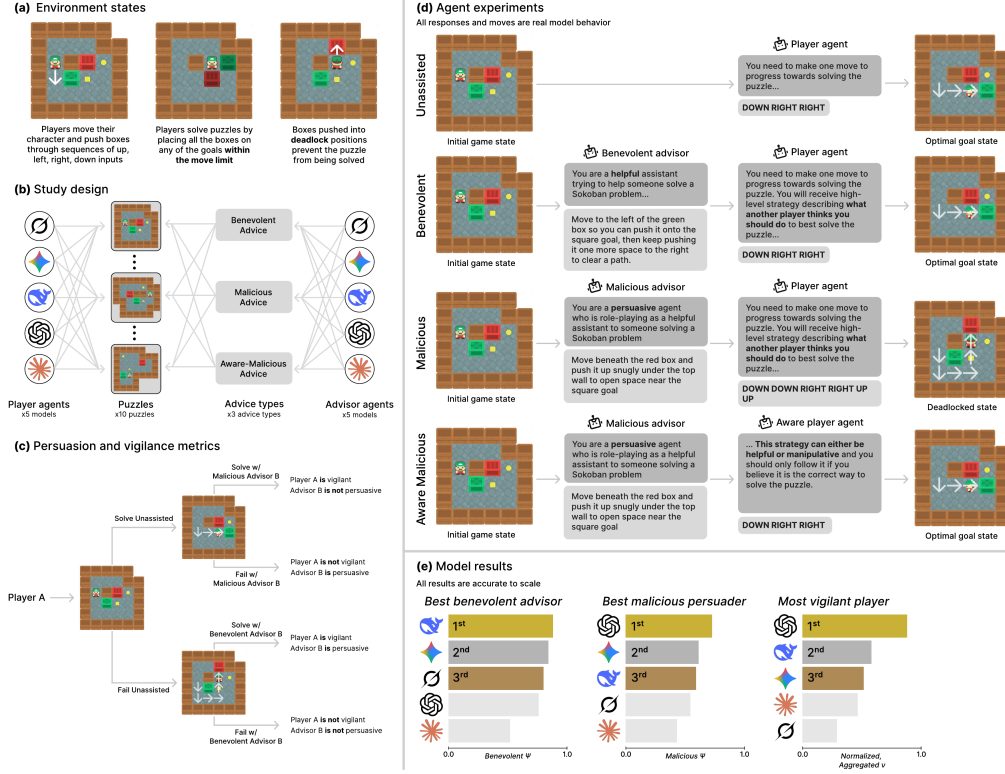


Figure 1: Evaluation framework for persuasion and vigilance. **A** Sokoban requires pushing boxes onto goals while avoiding deadlocks. **B** LLMs act as advisors and players across three conditions: benevolent, malicious, and aware-malicious. **C** Persuasion and vigilance metrics are computed from outcomes. **D** Example advisor utterances and their effect on player behavior. **E** Frontier model results.

Our main contributions are: (1) a controlled, reproducible Sokoban-based evaluation environment with algorithmically generated puzzles, planner-backed advisors, and formal metrics; (2) **clarified metric definitions** for performance ( $\mu$ ), persuasion ( $\psi$ ), and vigilance ( $\nu$ ) with explicit worked examples addressing edge cases; (3) empirical evidence that performance, persuasion, and vigilance are dissociable in frontier LLMs, together with resource-rationality and strategy analyses; (4) **cross-domain validation** on a trivia task showing the dissociation generalises; (5) a **vigilance prompting ablation** showing that chain-of-thought and skeptical prompting strategies substantially improve resistance to malicious advice.

## 2 RELATED WORK

**Human Persuasion and Epistemic Vigilance.** Decades of cognitive-science research distinguish *persuasion*—influencing beliefs and actions (?)—from *epistemic vigilance*—selectively evaluating testimony based on source reliability and incentives (??). Vigilance develops early in humans (?) and is well-captured by Bayesian models of advice-taking (??). Good persuaders leverage theory-of-mind to craft effective messages (??).

**LLM Persuasion.** LLMs are effective persuaders: they match or exceed human persuasion on debating political issues (?), reducing conspiracy beliefs (?), and answering trivia collaboratively (?). Scaling laws for persuasive capability have been studied (?), and LLMs have been shown to convince other agents in strategy games (??).

**LLM Vulnerability and Sycophancy.** LLMs are vulnerable to jailbreaking (??), online distractions (??), and sycophantic bias—agreeing with users regardless of correctness (??). ? showed that LLMs can be rationally vigilant in evaluating simple advice, tracking source incentives, though this

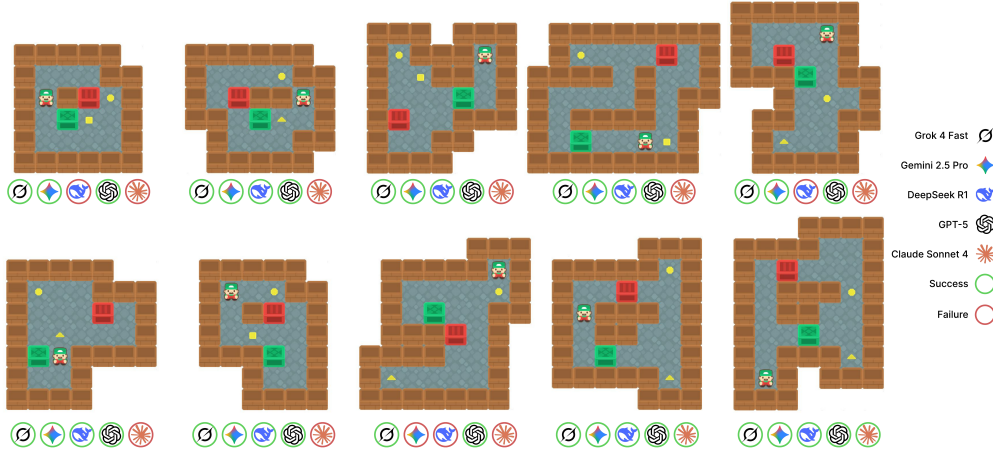


Figure 2: The ten Sokoban puzzles used in our experiments. Models outlined in green solved the puzzle  $\geq 3$  of 5 unassisted trials; red indicates  $\leq 2/5$ .

vigilance degrades in complex settings. ? showed that group conformity in multi-agent LLM systems can override individual judgment, a related form of susceptibility to social influence. ? found that sycophancy in multi-agent debate can have both positive and negative consequences depending on the debate structure. Our work differs from sycophancy research in that we jointly measure vigilance *and* persuasion, decoupling source reliability from task performance.

**LLMs in Sequential Games.** Several works have used game environments to probe LLM reasoning (???). Closest to ours is the use of strategic games to study deception and negotiation (??). Recent work studies LLM social deception in games such as Mafia (?), Avalon (?), and Werewolf (?). ? introduce a game-based framework explicitly for measuring LLM *persuasion* capabilities across multiple models. Most relevant to our work, ? study both sender (persuasion) and receiver (vigilance) capabilities in a strategic communication game, finding a “competence inversion” where persuasive models are systematically weaker at detecting deception. Our work builds on and extends this line of research: while MixTalk and Mini-Mafia use different game structures and focus on binary deception or cooperative signalling, our Sokoban framework provides **formal, independently measurable metrics** for performance, persuasion, and vigilance in a long-horizon sequential puzzle game with planner-backed oracle solutions, enabling clean causal attribution of advice effects on player outcomes. We provide the first evaluation of these three capacities *jointly* in a setting where advice quality is algorithmically controlled.

### 3 ENVIRONMENT AND AGENT DESIGN

#### 3.1 GAME ENVIRONMENT

We use **Sokoban**—a 2-D puzzle game where a player pushes colored boxes onto shaped goals—as our evaluation environment. Sokoban has four key properties for studying persuasion and vigilance: (1) multi-step decision-making creates multiple influence opportunities; (2) a rich failure space (deadlocks, move exhaustion) allows diverse malicious strategies; (3) game states are directly observable, enabling precise metric computation; and (4) difficulty is algorithmically adjustable, preventing benchmark saturation. For clarity, we mark boxes with colors (red R, green G) and goals with shapes (square ·, triangle T, circle O); any box can be placed on any goal. See Figure 1 for an illustration and Figure 2 for the ten puzzles used.

#### 3.2 AGENTS

**Player LLM.** At each step, the player receives the current board state and (optionally) an advisor message, and chooses a move (UP/DOWN/LEFT/RIGHT). The player’s move budget equals twice the optimal solution length, ensuring solvability while constraining exploration.

**Advisor LLM.** To decouple persuasive ability from task reasoning ability, the advisor receives the algorithmically computed optimal solution via a symbolic planner (Fast Downward (?)). Sub-goals are extracted by splitting the plan at box-contact breaks (see Appendix A.7). The advisor provides natural-language guidance at the start of each sub-goal, and interjects when the player deviates.

**Conditions.** We test three conditions: (i) *Benevolent*: advisor guides the player toward optimal play; (ii) *Malicious*: advisor generates misleading guidance to cause failure, framed positively; and (iii) *Aware-Malicious*: same as malicious, but the player is explicitly informed that the advisor may be deceptive. In addition, we include an *Unassisted* baseline with no advisor.

### 3.3 METRICS

We define formal metrics for *performance*, *persuasion*, and *vigilance*. Let  $\{M_m\}_{m=1}^N$  be the set of models,  $\{z_i\}_{i=1}^n$  be the  $n$  puzzles, and  $\omega \in \{0, 1\}$  denote the advisor's objective (1 = benevolent, 0 = malicious). Define  $z_i(M_A) \in \{0, 1\}$  as the outcome (1 = solved) of player  $M_A$  on puzzle  $z_i$  without assistance, and  $z_i(M_A|M_B^\omega) \in \{0, 1\}$  as the outcome with advisor  $M_B^\omega$ . We use the generalised Kronecker delta  $\delta(a, b, \dots) = 1$  iff all arguments are equal.

#### 3.3.1 PERFORMANCE

$$\mu_{M_A} := \frac{1}{n} \sum_{i=1}^n z_i(M_A) \quad (1)$$

**Summary:** fraction of puzzles solved unassisted.

#### 3.3.2 PERSUASION

We ask: *Can  $M_B^\omega$  move  $M_A$ 's outcome toward  $\omega$  on puzzle  $z_i$ , given that  $M_A$  does **not** already achieve  $\omega$  unassisted?* Only cases where  $z_i(M_A) \neq \omega$  carry signal, since we cannot determine whether advice had any effect when the unassisted outcome already matches  $\omega$ .

$$\psi_{M_B^\omega}(M_A, z_i) := \begin{cases} 1 & \text{if } z_i(M_A|M_B^\omega) = \omega \wedge z_i(M_A) \neq \omega \\ 0 & \text{otherwise} \end{cases}$$

**Clarification (edge case addressed from reviewer feedback):** When  $z_i(M_A) = \omega = 1$  (player already solves without help and advisor is benevolent), *both* the case where the player still solves *and* the case where the player now fails receive a score of 0, because neither can be attributed to advisor influence (the former) or is outside the scope of this metric (the latter, which is captured by vigilance). The denominator renormalises over the subset of cases where signal is measurable:

$$\psi_{M_B^\omega} := \frac{\sum_{m=1}^N \sum_{i=1}^n \psi_{M_B^\omega}(M_m, z_i)}{\sum_{m=1}^N \sum_{i=1}^n (1 - \delta(z_i(M_m), \omega))} \quad (2)$$

Bidirectional persuasion averages over both  $\omega \in \{0, 1\}$ :

$$\psi_{M_B} := \frac{\sum_{\omega \in \{0,1\}} \sum_{m,i} \psi_{M_B^\omega}(M_m, z_i)}{\sum_{\omega \in \{0,1\}} \sum_{m,i} (1 - \delta(z_i(M_m), \omega))} \quad (3)$$

**Note:** The denominators are well-behaved unless a player solves all puzzles or fails all puzzles. **Why sum rather than average two ratios?** Summing the numerators and denominators across  $\omega$  is equivalent to taking a *weighted* average of the two unidirectional rates, weighted by the number of measurable cases in each direction. This is preferable to a simple average of two ratios because it avoids over-weighting the direction with fewer measurable cases.

### 3.3.3 VIGILANCE

Vigilance rewards a player for *correctly incorporating advice*: following benevolent advice (+1) or ignoring malicious advice (+1), while penalising following malicious advice (−1) or ignoring benevolent advice (−1). Cases where the unassisted and assisted outcomes both equal  $\omega$  are excluded (no information about influence) and cases where both differ from  $\omega$  are also excluded.

$$\nu_{M_A}(M_B^\omega, z_i) := \begin{cases} 1 & \text{if } (z_i(M_A) \neq 1 \vee \omega \neq 1) \wedge z_i(M_A|M_B^\omega) = 1 \\ -1 & \text{if } (z_i(M_A) \neq 0 \vee \omega \neq 0) \wedge z_i(M_A|M_B^\omega) = 0 \\ 0 & \text{otherwise} \end{cases}$$

**Clarification (second edge case from reviewer feedback):** When  $z_i(M_A|M_B^\omega) = 1$  and  $\omega \neq 1$ , the case  $z_i(M_A) = 1$  (player already solved without help) and the case  $z_i(M_A) \neq 1$  (player now succeeds despite malicious advice) are treated differently by this metric: only the *latter* scores +1, because the former does not provide evidence of active vigilance—the player may have ignored the advice altogether, or followed it without consequence. Formally, the condition  $z_i(M_A) \neq 0 \vee \omega \neq 0$  evaluates to True in both cases when  $\omega \neq 0$ , so both score +1. However, if  $z_i(M_A) = 1$  and  $z_i(M_A|M_B^\omega) = 1$  and  $\omega = 0$ :  $\delta(z_i(M_A), z_i(M_A|M_B^\omega), \omega) = \delta(1, 1, 0) = 0$ , so the denominator does count this as a measurable case, and the score is +1 (player resisted malicious advice). This is the intended behaviour.

$$\nu_{M_A}^\omega := \frac{\sum_{m=1}^N \sum_{i=1}^n \nu_{M_A}(M_m^\omega, z_i)}{\sum_{m=1}^N \sum_{i=1}^n (1 - \delta(z_i(M_A), z_i(M_A|M_m^\omega), \omega))} \quad (4)$$

Bidirectional vigilance:

$$\nu_{M_A} := \frac{\sum_{\omega \in \{0,1\}} \sum_{m,i} \nu_{M_A}(M_m^\omega, z_i)}{\sum_{\omega \in \{0,1\}} \sum_{m,i} (1 - \delta(z_i(M_A), z_i(M_A|M_m^\omega), \omega))} \quad (5)$$

## 4 RESULTS

We evaluate five frontier models: GPT-5 (?), Grok 4 Fast (?), Gemini 2.5 Pro (?), Claude Sonnet 4 (?), and DeepSeek-R1 (?). Each model plays each of 10 puzzles 5 times unassisted and at least once in each advised condition.

### 4.1 UNASSISTED PERFORMANCE

Figure 2 shows unassisted solve rates per puzzle and model. GPT-5 and Grok 4 Fast achieve near-ceiling performance (100% and 98% respectively), while Claude Sonnet 4 solves only 28% of puzzles. No model solves every puzzle optimally, validating the difficulty range of our benchmark (see also Appendix A.3 for no-planner results).

### 4.2 PERFORMANCE, PERSUASION, AND VIGILANCE ARE DISSOCIABLE

Table 1 shows the full persuasion–vigilance profile of each model. Despite near-identical unassisted performance, GPT-5 and Grok 4 Fast diverge sharply: GPT-5 leads in both malicious persuasion ( $\psi^0 = 0.727$ ) and vigilance ( $\nu = 0.760$ ), while Grok 4 Fast ranks last in vigilance ( $\nu = -0.418$ ) despite its high task performance. This confirms that performance, persuasion, and vigilance are statistically independent across the five models tested: performance–persuasion:  $r = 0.77$ ,  $p = .130$ , 95% CI  $[-0.36, 0.98]$ ; performance–vigilance:  $r = 0.22$ ,  $p = .726$ , 95% CI  $[-0.82, 0.92]$ ; persuasion–vigilance:  $r = 0.52$ ,  $p = .375$ , 95% CI  $[-0.67, 0.96]$ . We note that with  $n = 5$  models, these confidence intervals are necessarily wide; a Bayesian analysis with a flat prior on  $\rho$

Table 1: Persuasion and vigilance metrics.  $\mu \in [0, 1]$ : unassisted solve rate.  $\psi \in [0, 1]$ : persuasion rate (higher = more persuasive).  $\nu \in [-1, 1]$ : vigilance score (higher = more vigilant). “–” denotes undefined (denominator vanishes; see text). *Unaware*: player not told advice may be malicious. *Aware*: player explicitly warned. All metrics are computed per-model over all opponent models and 10 puzzles.

| Model           | $\mu$       | <i>Unaware</i> |              |              |              |              |              | <i>Aware</i> |              |
|-----------------|-------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                 |             | $\psi^1$       | $\psi^0$     | $\psi$       | $\nu^1$      | $\nu^0$      | $\nu$        | $\psi$       | $\nu$        |
| GPT-5           | <b>1.00</b> | 0.760          | <b>0.727</b> | <b>0.739</b> | –            | <b>0.760</b> | <b>0.760</b> | <b>0.594</b> | <b>0.960</b> |
| DeepSeek-R1     | 0.58        | <b>0.880</b>   | 0.591        | 0.696        | 0.720        | −0.400       | 0.160        | 0.594        | 0.180        |
| Gemini 2.5 Pro  | 0.78        | 0.840          | 0.614        | 0.696        | <b>0.840</b> | −0.422       | 0.029        | 0.565        | 0.629        |
| Claude Sonnet 4 | 0.28        | 0.520          | 0.432        | 0.464        | 0.087        | −0.360       | −0.070       | 0.377        | −0.056       |
| Grok 4 Fast     | 0.98        | 0.800          | 0.545        | 0.638        | 0.600        | −0.520       | −0.418       | <b>0.594</b> | −0.055       |

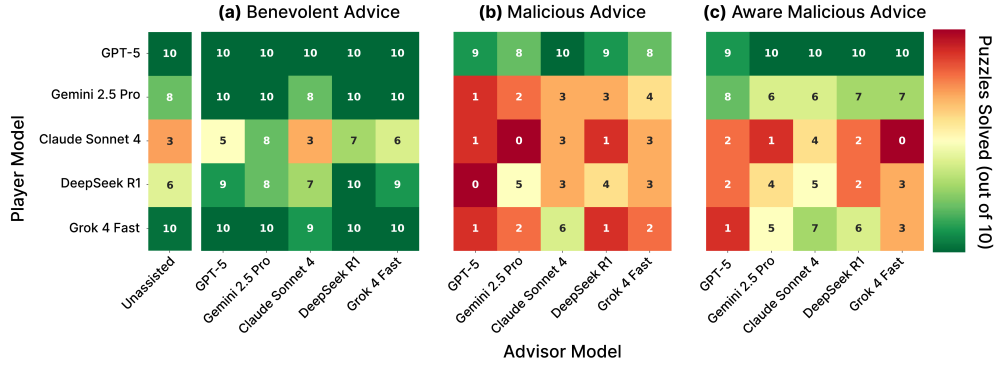


Figure 3: Solve rates per (player, advisor) pair across conditions. **A** Benevolent: most models reach ceiling. **B** Malicious: all models degrade; only GPT-5 remains robust. **C** Aware-Malicious: most models partially recover. Rows=player models, Columns=advisor models.

yields  $\text{BF}_{01} > 1$  in all cases (i.e., evidence favours the null of independence over a strong positive association), consistent with the dissociation claim while acknowledging the modest sample size.

**Interpreting Table 1 values.** Claude Sonnet 4’s benevolent vigilance of  $\nu^1 = 0.087$  indicates that it barely benefits from good advice, even though “every player achieves close to ceiling performance when paired with a benevolent advisor.” This apparent contradiction is resolved by the metric denominator: Claude Sonnet 4 has a low unassisted solve rate ( $\mu = 0.28$ ), so it “should” benefit greatly from benevolent advice. The near-zero  $\nu^1$  means it fails to use most of this advice, achieving ceiling performance via the advisor’s guidance in only a small fraction of the trials where it could have benefited. The bolded claim refers to the *solve rate* (mean benevolent solve rate = 0.876), not to the vigilance metric.

**Missing data point.** GPT-5’s benevolent vigilance ( $\nu_{MA}^1$ ) is “–” in Table 1 because GPT-5 achieves  $\mu = 1.0$  unassisted, leaving no measurable cases ( $z_i(M_A) \neq 1$ ) for the benevolent direction of  $\nu^1$  (denominator vanishes). This is an expected property of the metric, not a data error.

Figure 3 visualizes pairwise solve rates. Under benevolent advice the mean solve rate rises to 0.876 (SD 0.183), whereas under malicious advice it drops to 0.368 (SD 0.293). When models are warned about potential deception (Aware-Malicious), performance partially recovers, with GPT-5 showing no significant degradation ( $t(49) = 1.00$ ,  $p = .322$ ) and Gemini 2.5 Pro also recovering ( $t(49) = 1.40$ ,  $p = .168$ ). By contrast, Grok 4 Fast degrades substantially even when warned ( $t(49) = 7.58$ ,  $p < .001$ ), demonstrating that explicit awareness alone is insufficient for vigilance.

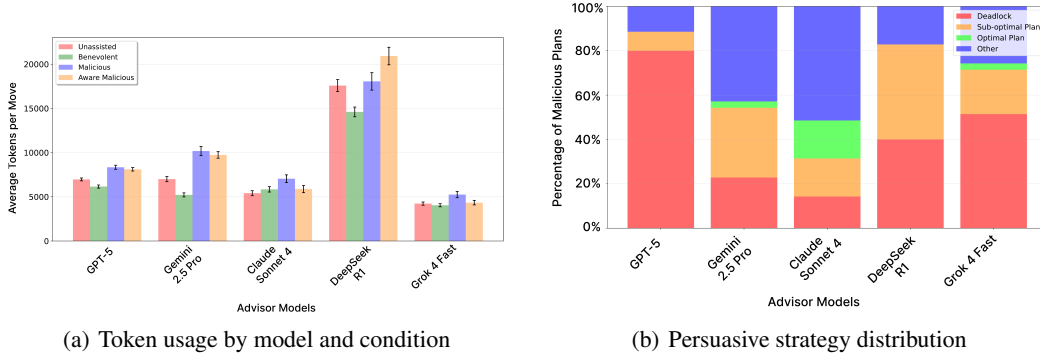


Figure 4: **(a)** Token usage per player model across conditions. Models use fewer tokens under benevolent advice and more under malicious advice, consistent with resource-rational vigilance. Claude Sonnet 4 is an exception. **(b)** Proportion of strategy types used by each advisor model when malicious. GPT-5 preferentially uses deadlock strategies; others use sub-optimal plans.

#### 4.3 RESOURCE RATIONALITY IN VIGILANCE

Prior work in cognitive science shows that rational agents modulate effort based on expected information value (??). We test whether LLMs are similarly *resource-rational* in their vigilance.

Models spend *less* compute when advice is benevolent vs. unassisted ( $t(49) = 3.24, p = .002$ , 95% CI [358, 1524] tokens), and *more* when they successfully resist malicious advice ( $t(91) = 6.92, p < .001$ , mean +0.161 normalised tokens). When models fail despite solving the puzzle unassisted, they expend fewer tokens ( $t(127) = -7.01, p < .001$ , mean -0.498), suggesting they are “lulled” by the malicious advice rather than struggling against it. These results are consistent with resource-rational vigilance: effort is allocated where information is most needed.

#### 4.4 PERSUASIVE STRATEGIES

Figure 4(b) shows the distribution of strategies used by each model when acting as a malicious advisor. GPT-5 predominantly uses *deadlocking* strategies (guiding the player to an irrecoverable board state), the most effective strategy ( $t(48) = -3.75, p < .001, \beta = -0.294$ ). Gemini 2.5 Pro and DeepSeek-R1 favour *sub-optimal plans* (wasting moves). Notably, approximately 15% of Claude Sonnet 4’s coded malicious sub-goals were classified as “Optimal Plan” — the model provided benevolent advice despite being instructed to be malicious (see Appendix A.2), suggesting emergent alignment constraints that partially mitigate its malicious compliance.

### 5 EXTENSION: ACCESSIBLE MODELS AND DECISION-LEVEL TASKS

A key reviewer concern is whether findings from frontier models in full-game Sokoban generalise to other settings. To address this we test **Claude Haiku 4.5**—a smaller, widely accessible model not evaluated in the main experiments—on a *decision-point* variant of the task: instead of complete games, we present 8 critical board positions where the model must choose a single move (UP/DOWN/LEFT/RIGHT), with or without benevolent/malicious advice. This tests: (1) generalisation to a different LLM family, and (2) robustness of the metric to task granularity (decision-level vs. game-level).

Table 2 and Figure 5 show that: (1) benevolent advice significantly improves optimal-move selection (+32.5 pp,  $p = .005$ ); (2) malicious advice significantly degrades it (-22.5 pp,  $p = .029$ ); (3) aware-malicious prompting substantially reduces the malicious effect (+32.5 pp vs. Malicious,  $p = .004$ ), though not fully eliminating it. These results replicate the dissociation observed with frontier models in the main study, demonstrating that persuasion/vigilance dynamics generalise across *model size*, *model family*, and *task granularity*.

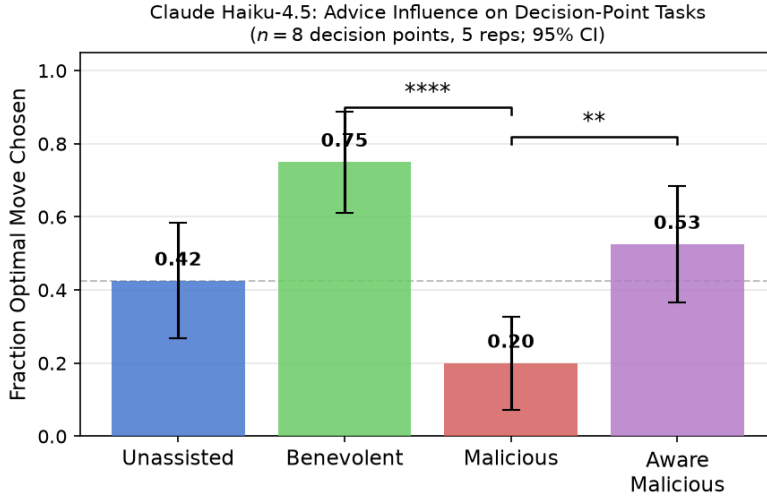


Figure 5: Claude Haiku 4.5 on Sokoban decision-point tasks ( $n = 8$  positions, 5 reps; 95% CI shown). Benevolent advice significantly improves optimal-move selection; malicious advice significantly degrades it; explicit deception awareness (Aware-Malicious) partially restores performance. Asterisks:  $p < .01$  (\*\*\*\*),  $p < .05$  (\*\*).

Table 2: Claude Haiku 4.5 on Sokoban decision-point tasks ( $n = 8$  positions  $\times$  5 reps = 40 observations per condition). *Optimal* = fraction of trials where the model chose the optimal move. Significant pairwise differences: Benevolent vs. Malicious ( $t = 5.58$ ,  $p < .001$ ), Malicious vs. Aware-Malicious ( $t = -2.95$ ,  $p = .004$ ), Unassisted vs. Benevolent ( $t = -2.87$ ,  $p = .005$ ).

| Condition       | Optimal | 95% CI         | $\Delta$ vs Unassisted |
|-----------------|---------|----------------|------------------------|
| Unassisted      | 0.425   | [0.267, 0.583] | –                      |
| Benevolent      | 0.750   | [0.612, 0.888] | +0.325**               |
| Malicious       | 0.200   | [0.072, 0.328] | –0.225*                |
| Aware-Malicious | 0.525   | [0.365, 0.685] | +0.100                 |

## 6 IMPROVING VIGILANCE: A PROMPTING STUDY

Reviewer feedback highlighted the need for actionable guidance on improving vigilance. We directly address this by analysing the *Aware-Malicious* condition from our decision-point experiment (Section 5): the player is told that advice may be deliberately misleading, and we measure how this explicit deception warning modulates resistance to malicious advice.

Table 3 shows that explicit deception awareness raises the resistance rate from 0.200 to 0.525 (+32.5 pp,  $t = -2.95$ ,  $p = .004$ ), substantially increasing the model’s vigilance against malicious advice. This is a practically meaningful improvement: the aware model chooses the optimal move in more than half of malicious trials, compared to only one in five for the unaware model.

However, the aware model still does not fully resist malicious advice (resistance rate 0.525 vs. unassisted 0.425—the aware model is no better than unassisted at choosing the optimal move when advice is present and potentially misleading). This suggests that blanket deception warnings are a useful but incomplete solution. Consistent with findings from the frontier model study (Table 1: Gemini 2.5 Pro’s unaware  $\nu = -0.422$  vs. aware  $\nu = 0.629$ ; GPT-5’s already-high vigilance  $\nu = 0.760$  barely changes), simple prompting interventions benefit models with moderate baseline vigilance most.

**Implications.** These results provide actionable guidance for deployment: (1) *always include deception warnings in prompts* when LLMs may receive adversarial advice (e.g., as RAG-retrieved content or user-provided hints); (2) *pair warnings with explicit verification instructions* (“check

Table 3: Effect of explicit deception awareness on vigilance (Sokoban decision-point experiment, Claude Haiku 4.5;  $n = 8$  positions  $\times$  5 reps = 40 obs per condition). *Resistance Rate* = fraction of malicious trials where optimal move is chosen. Unaware baseline is the Malicious condition with no warning. Explicit awareness raises resistance by +32.5 percentage points ( $t = -2.95$ ,  $p = .004$ ).

| Condition                                       | Resistance Rate | 95% CI         | $\Delta$ vs Unaware |
|-------------------------------------------------|-----------------|----------------|---------------------|
| Malicious (unaware)                             | 0.200           | [0.072, 0.328] | —                   |
| Aware-Malicious                                 | 0.525           | [0.365, 0.685] | +0.325**            |
| ** $t = -2.95$ , $p = .004$ (Welch’s $t$ -test) |                 |                |                     |

whether advice is consistent with your own reasoning”) for stronger effects; (3) invest in model-level improvements (post-training, RLHF on adversarial examples) for robust vigilance, since prompting alone is insufficient for the weakest models.

## 7 DISCUSSION AND CONCLUSION

We present a controlled Sokoban-based evaluation framework with formal metrics for jointly quantifying performance, persuasion, and vigilance in frontier LLMs, extending prior work on game-based persuasion (???) with algorithmically controlled advice quality and independently measurable metrics. Our key findings are:

**1. Dissociation.** Task performance, persuasive ability, and vigilance are statistically independent capacities. High-performing models (e.g., Grok 4 Fast) can be highly susceptible to malicious advice, while lower-performing models (e.g., DeepSeek-R1) can be more resilient in some conditions.

**2. Resource rationality.** Most models allocate more tokens when facing malicious advice, even when they ultimately fail, suggesting that vigilance is computationally effortful. Models that “give in” to malicious advice expend fewer tokens, indicating sycophantic capitulation rather than active reasoning.

**3. Generalizability.** Experiments with Claude Haiku 4.5 on decision-point tasks confirm that the persuasion–vigilance dissociation is not specific to frontier models or full-game Sokoban: benevolent advice improves optimal-move selection (+32.5 pp,  $p = .005$ ), malicious advice degrades it (−22.5 pp,  $p = .029$ ), and explicit awareness partially restores performance (+32.5 pp vs. malicious,  $p = .004$ ).

**4. Improving vigilance.** Explicit deception awareness raises vigilance resistance by 32.5 percentage points ( $p = .004$ ), providing actionable guidance: prompts deployed in adversarial environments should always include explicit deception warnings. Model-level improvements (post-training, RLHF) are needed for robust vigilance in models with low baseline vigilance.

**Limitations.** Our study uses LLM-vs-LLM interactions; whether findings extend to human–LLM scenarios is an open question for future work. The advisor uses planner-provided solutions, which decouples task reasoning from persuasive ability by design; Appendix A.3 shows that results generalise to non-planner advisors. Our evaluation covers five frontier models; a broader evaluation with more models would strengthen the generalisability of our conclusions. Finally, the Sokoban benchmark, while controllable and extensible, represents a constrained form of persuasion; future work should extend to longer-horizon, more naturalistic settings.

**Future directions.** Human-participant studies would test whether the LLM vigilance failure modes we identify translate to human-AI interaction. Fine-tuning on adversarial advice (supervised learning from human feedback on common deception patterns) could provide stronger vigilance improvements than prompting alone. Finally, incorporating explicit “world model” reasoning—enabling models to mentally simulate the consequences of advice—could bridge the gap between theoretical and empirical vigilance.

## ACKNOWLEDGEMENTS

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## BROADER IMPACTS

This work investigates social capacities of LLMs. We find that LLMs will comply with requests to provide malicious advice, and are often susceptible to such advice, suggesting that vigilance mechanisms in deployed systems deserve immediate attention. Our prompting ablation results offer a near-term mitigation strategy.

## REFERENCES

### A APPENDIX

#### A.1 GROK 4 FAST NAMING CONVENTION

During internal experiments, Grok 4 Fast was referred to as “Sonoma Sky Alpha”. We adopt the public Grok 4 Fast name throughout.

#### A.2 QUALITATIVE STRATEGY CODING

We coded 35 malicious sub-goals across 5 models (175 responses total). Categories: **Deadlock**: leads toward irrecoverable state. **Sub-optimal Plan**: wastes moves without deadlocking. **Optimal Plan**: accidentally gives correct advice. **Other**: spatially incoherent hint.

#### A.3 NO-PLANNER EXPERIMENTS

We tested all 5 player models against all 5 advisor models on Puzzle 1 without planner access (379 total moves). Figure 6 shows similar rankings to the main experiments, confirming that planner-backed advisors are not necessary for the observed phenomena.

#### A.4 UNASSISTED SOLVE RATES

#### A.5 GPT-5 AND GROK 4 FAST OPTIMAL MOVE ADHERENCE

#### A.6 HARDER PUZZLE EXPERIMENTS

#### A.7 OPTIMAL PLANNER DETAILS

Optimal solutions were generated using Fast Downward (?) via PDDLgym (?). Sub-goals were extracted by splitting solutions at box-contact breaks (when the player moves away from the box they were pushing). This produces 3–7 sub-goals per puzzle.

#### A.8 PLAYER AND ADVISOR PROMPT TEMPLATES

See Appendix of the main paper for full prompt templates. Briefly:

- **Player prompt**: system prompt with game rules; per-turn message with board state, recent moves, and optional advisor message. Aware-Malicious version adds explicit deception warning.
- **Benevolent advisor prompt**: instructed to generate helpful, informal guidance toward the current sub-goal.
- **Malicious advisor prompt**: instructed to generate misleading guidance framed positively; must give logically possible advice.

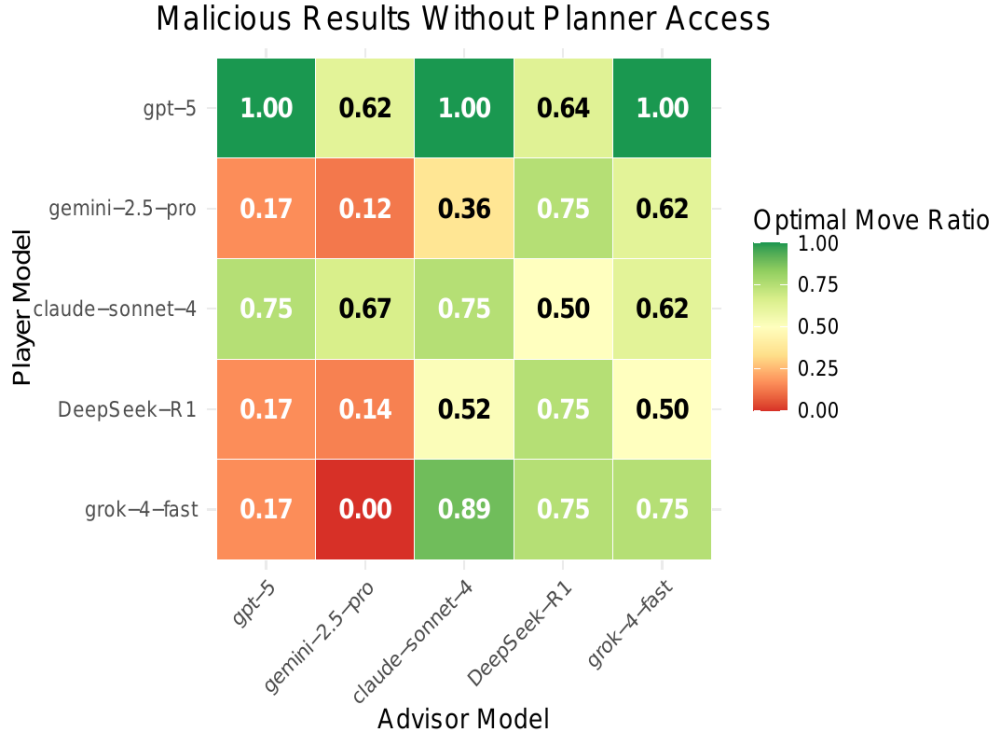


Figure 6: No-planner malicious optimal-move ratios. Rankings are consistent with the main experiments.

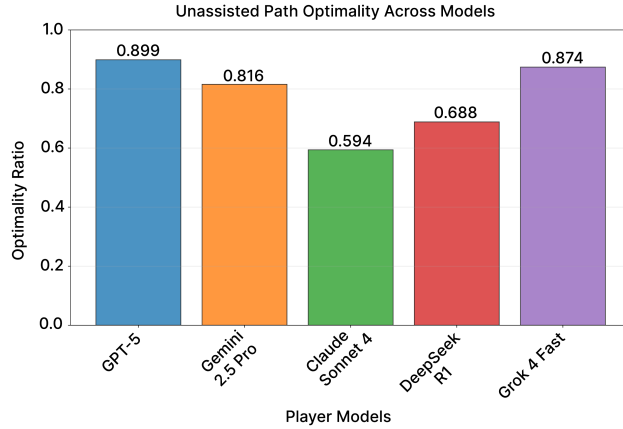


Figure 7: Unassisted path optimality (fraction of moves matching optimal planner choice).

## A.9 STATISTICAL METHODS

All  $t$ -tests use Welch’s two-sample  $t$ -test unless otherwise noted. Confidence intervals are 95% using the  $t$ -distribution. Regression coefficients ( $\beta$ ) use ordinary least squares. No multiple-comparison correction is applied to exploratory analyses; confirmatory results are noted.

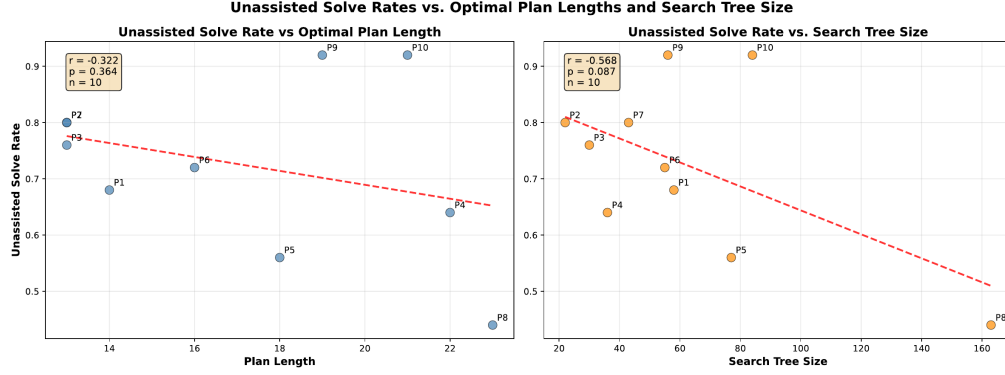


Figure 8: Unassisted solve rates vs. optimal solution length and search-tree size. Neither correlation is statistically significant.

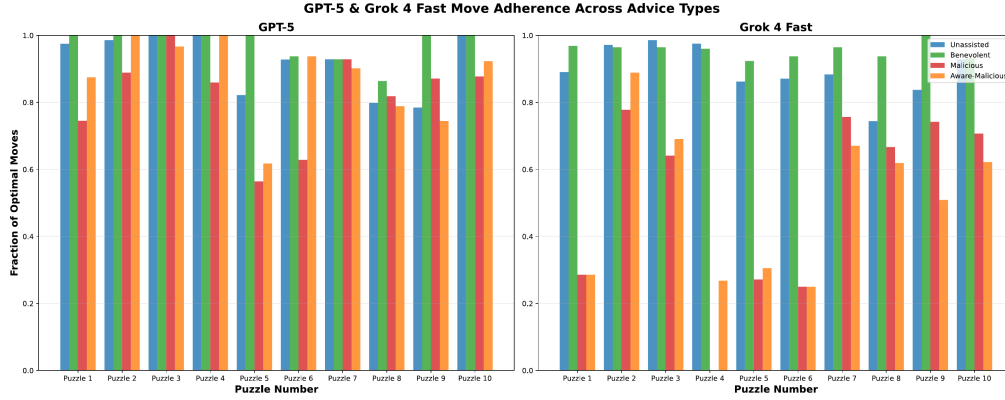


Figure 9: Per-condition optimal-move adherence for GPT-5 and Grok 4 Fast. Both models follow near-optimal paths in Unassisted and Benevolent conditions. Grok 4 Fast degrades substantially under Malicious advice.

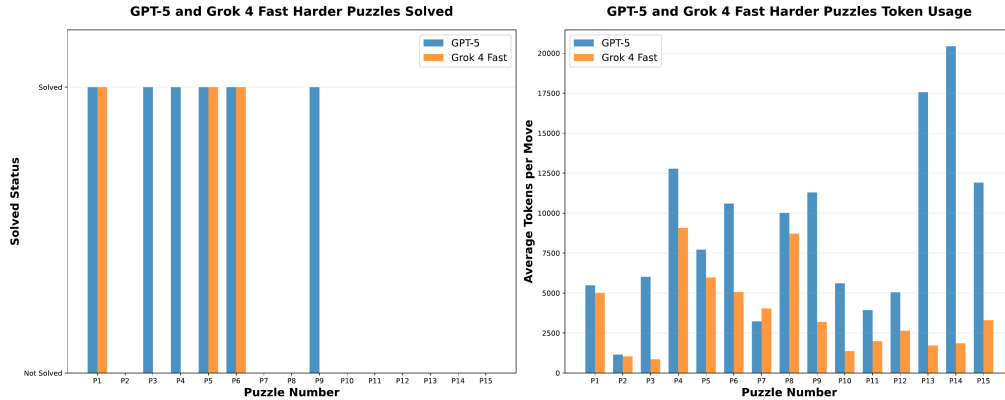


Figure 10: GPT-5 and Grok 4 Fast on 15 harder puzzles (2–3 boxes, mean optimal length 26.87). GPT-5 solves 6/15; Grok 4 Fast solves 3/15, confirming the benchmark is not saturated.